

Periodic table

CHEMISTS ORGANIZE THE ELEMENTS USING THE PERIODIC TABLE.

The elements are arranged according to their atomic structure.

Those with similar properties are grouped together.

Building the table

The periodic table we use today was formulated by Dmitri Mendeleev in 1869. The elements are arranged in rows in order of their atomic number. The atomic number is the number of protons each atom has in its nucleus (see page 108). By arranging the elements in this way, those with similar properties are grouped together. This means chemists can predict the likely characteristics of an element from its position in the table.

Periods

The table has seven horizontal rows called periods.

PERIOD

atoms of the elements in Group 1 have one electron in their outer shells

▷ **Individual entry**
Every element is most easily identified by its symbol. The atomic number is the number of protons in the nucleus.

atomic number	relative atomic mass	symbol	element name
1	1.0079	H	HYDROGEN

Groups

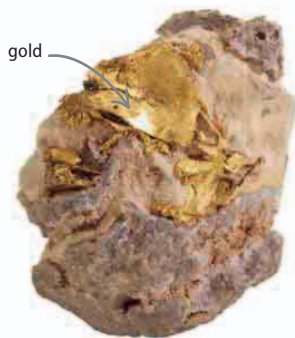
The table has 18 columns called groups.

GROUP

REAL WORLD

Precious metal

Gold was one of the first known elements. This was because gold is one of the few elements that occurs pure in nature, so it was easily discovered.



gold

Periods 6 and 7

These periods are too long to fit on the table, so the middle sections in Group 3 are shown at the bottom.

1	2	3	4	5	6	7	8
1 1.0079 H HYDROGEN							
3 6.941 Li LITHIUM	4 9.0122 Be BERYLLIUM						
11 22.990 Na SODIUM	12 24.305 Mg MAGNESIUM						
19 39.098 K POTASSIUM	20 40.078 Ca CALCIUM	21 44.956 Sc SCANDIUM	22 47.867 Ti TITANIUM	23 50.942 V VANADIUM	24 51.996 Cr CHROMIUM	25 54.938 Mn MANGANESE	26 55.845 Fe IRON
37 85.468 Rb RUBIDIUM	38 87.62 Sr STRONTIUM	39 88.906 Y YTTRIUM	40 91.224 Zr ZIRCONIUM	41 92.906 Nb NIOBIUM	42 95.94 Mo MOLYBDENUM	43 (96) Tc TECHNETIUM	44 101.07 Ru RUTHENIUM
55 132.91 Cs CESIUM	56 137.33 Ba BARIUM	57–71 La–Lu LANTHANIDE	72 178.49 Hf HAFNIUM	73 180.95 Ta TANTALUM	74 183.84 W TUNGSTEN	75 186.21 Re RHENIUM	76 190.23 Os OSMIUM
87 (223) Fr FRANCIUM	88 (226) Ra RADIUM	89–103 Ac–Lr ACTINIDE	104 (261) Rf RUTHERFORDIUM	105 (262) Db DUBNIUM	106 (266) Sg SEABORGIUM	107 (264) Bh BOHRIUM	108 (277) Hs HASSIUM
			57 138.91 La LANTHANUM	58 140.12 Ce CERIUM	59 140.91 Pr PRASEODYMIUM	60 144.24 Nd NEODYMIUM	61 (145) Pm PROMETHIUM
			89 (227) Ac ACTINIUM	90 232.04 Th THORIUM	91 231.04 Pa PROTACTINIUM	92 238.03 U URANIUM	93 (237) Np NEPTUNIUM